# srimpro Library

This is a Python library that contains many useful functions for automated data extracting, calculating, and plotting for SRIM simulation post-processing. These functions use the .txt output files that can be obtained from SRIM as described below. This document contains detailed instructions on what each of these functions do, how to use them, and explains their various settings/inputs.

## Contents

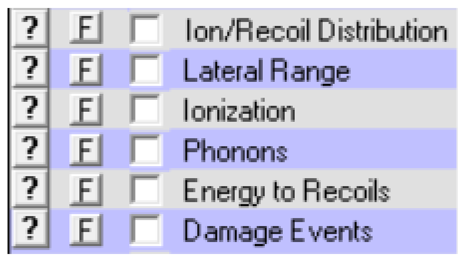
# Using srimpro in Python

After installing and setting up srimpro, all functions can be accessed by including the line *import srimpro.functions as pro* (can also be similarly imported as any other title) at the top of a .py file. Then, the functions can be used by including “pro.” at the beginning of any function’s name. For example, to use rangeAndDpa, the function call would be *pro.rangeAndDpa()*.

# All Functions

All functions in this library have the following common inputs:

- path: The file path to the folder containing the SRIM output files (.txt files).
    - This folder should contain all the SRIM output files shown in the image below no matter what function you are trying to use, since some functions use unexpected files.




    - This folder should also include the TDATA.txt file that SRIM automatically generates when downloading any of the above files.
    - The path should look something like *“/Users/noahm/OneDrive/Documents/Work/RE-MAT CERC Group 2026/SRIM/1.5 MeV Si in ZrN and Si/Output”.* Note that if copy your folder as path from file explorer, you may have to replace backslashes with forward slashes and remove the “C:” that appears before the file path (if using Windows).
- x1: The minimum depth value to use for calculations and/or plots. Optional, defaults to 0.
    - This value must be given in angstrom; functions will not work properly if x1 is given in any other units.
- x2: The maximum depth value to use for calculations and/or plots. Optional, defaults to the maximum depth used in the SRIM simulation(s).
    - This value must be given in angstrom; functions will not work properly if x2 is given in any other units.

All functions in this library make the following assumptions:

- path given as a string
- The data in all SRIM output files is given in units with angstrom and eV, other than density values which use cm

Note that all functions within this library are compatible with both single-layer and multilayer target materials and can deal with calculations for all target materials automatically. However, additional care should be taken when performing certain calculations on multilayer targets due to inaccurate recording of layer depth information by SRIM. This is done using the optional layerdepth input, explained below for each affected function.

# Integration Functions

These functions automatically extract data from the SRIM output files, then integrates between specified depth limits using Simpson's rule (more accurate than trapezoid rule) to get the total values (i.e. total number of vacancies, total ionization energy deposited, etc.) between the specified limits. The calculated values are then returned by the function (not automatically printed to console).

## totDisplacements

### Description

This function calculates the total number of vacancies and total number of replacements between the specified depth values (average per ion). Number of vacancies and number of replacements are returned in that order.

### Unique Inputs

None.

## totEnergyDep

### Description

This function calculates the total amount of energy deposited as ionization energy and damage energy, as well as the percentages for each, between the specified depth values (average per ion). Damage energy can be calculated using Phonon Method 1, Phonon Method 2 or Energy Conservation Method (all described below). Two 1D NumPy arrays are returned, with two elements each. The elements of the first array are the energy lost as ionization energy and the percent of energy lost as ionization energy in that order. The elements of the second array are the energy lost as damage energy and the percent of energy lost as ionization energy in that order.

### Unique Inputs

- method: what method to use to calculate damage energy. Optional, defaults to 'cons'.
    - 'phonons_1': damage energy is calculated by summing *phonons by ion* and *phonons by recoils* from the PHONON.txt file
    - 'phonons_2': damage energy calculated using only *phonons by recoils* from the PHONON.txt file
    - 'cons': damage energy calculated by subtracting the total ionization energy deposited between x1 and x2 from the total incident ion energy deposited between x1 and x2

## totStopping

### Description

This function calculates the total amount of incident ion energy lost to nuclear and electronic stopping between the specified depth values (average per ion). Two 1D NumPy arrays are returned, with two elements each. The elements of the first array are the energy lost to electronic stopping and the percent of energy lost as electronic stopping in that order. The elements of the second array are the energy lost to nuclear stopping and the percent of energy lost to nuclear stopping in that order.

### Unique Inputs

None.

# Basic Plotting Functions

These functions automatically extract data from the SRIM output files, then perform calculations and generate commonly used, publication-quality plots or export calculated data to excel. These functions have various optional inputs, and the plotting parameters (i.e. line width, colours, etc.) can be easily edited within the functions to allow for plot customization.

Common inputs:

- output: what output to get from the function. Optional, defaults to 'plot'
  - 'plot': function generates a plot
  - 'excel': function exports the calculated data to excel

## rangeAndDpa

### Description

This function calculates damage dose (even for multilayer targets) and generates either a dual y-axis plot of the ion distribution and damage dose, or a single y-axis plot of one of these profiles. Ion distribution can be plotted in units of [at. %] or [(Atoms/cm^3)/(Atoms/cm^2)].

### Unique Inputs

- fluence: the fluence to use for the damage dose calculations. Must be given in $cm^{-2}$.
- units: what units to use for ion density. Optional, defaults to 'at_%'
  - 'at_%': use ion density units of [at. %]
  - 'density_per_fluence': use ion density units of [(Atoms/cm^3)/(Atoms/cm^2)]
- ion_range: indicator for plotting the ion distribution profile. Optional, defaults to True.
  - True: enable this option
  - False: disable this option
- dose: indicator for plotting the damage dose profile. Optional, defaults to True.
  - Same options as ion_range

- layerdepth: an array of the bottom depths of each layer of the target material, must be given in angstrom. Optional, if not specified the layer bottom depths will be automatically read from the TDATA.txt file. layerdepth should always be manually input for multilayer targets to avoid calculation errors but does not need to be specified for single layer targets. Note that if specified, layerdepth must be an array, even if there is only 1 target material layer (must be a 1-element array in this case).

## Unique Assumptions

- fluence is given as a scalar

## Example Output

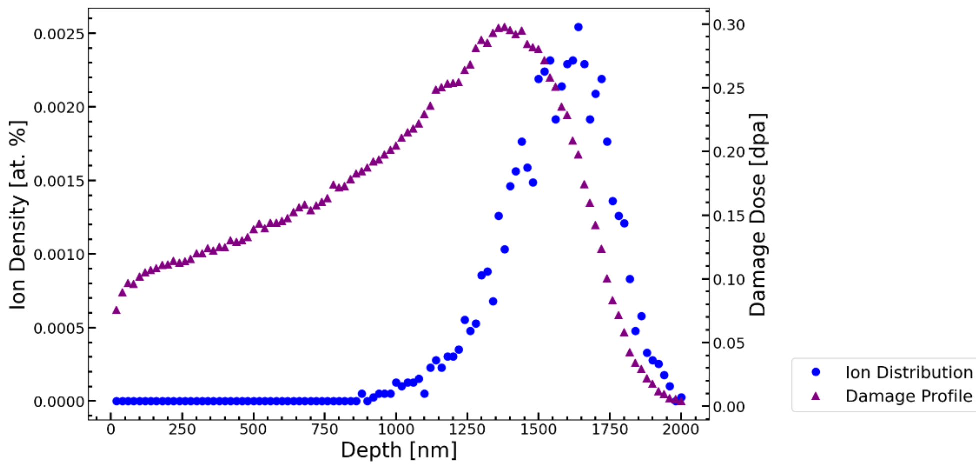



*Figure 1: Ion distribution and damage dose plot for 10 MeV Au in SiC. 2000 ions calculated, fluence = 1e14.*

# energyDep

## Description

This function generates a plot of energy the deposition profiles. Any combination of damage energy, total ionization energy, ionization by ion, and ionization by recoil can be plotted. The damage energy profile can be calculated using Phonon Method 1, Phonon Method 2, or Energy Conservation Method (all described below). The y-axis can use a linear or logarithmic scale (base 10).

## Unique Inputs

- damage: indicator for plotting damage energy. Optional, defaults to True.
  - True: enable this option
  - False: do not enable this option
- ionizTot: indicator for plotting total ionization energy. Optional, defaults to True.
  - Same options as damage
- ionizIon: indicator for plotting ionization energy by ion. Optional, defaults to True.
  - Same options as damage
- ionizRec: indicator for plotting ionization energy by recoils. Optional, defaults to True.

- Same options as damage

- method: what method to use to calculate damage energy. Optional, defaults to 'cons'.
  - 'phonons_1': damage energy profile is calculated by summing *phonons by ion* and *phonons by recoils* from the PHONON.txt file
  - 'phonons_2': damage energy calculated using only *phonons by recoils* from the PHONON.txt file
  - 'cons': damage energy profile is calculated by multiplying by multiplying each layer of the total displacement profile by a local Tdam/v factor specific to that layer, where Tdam is the damage energy calculated as incident ion energy deposited in the layer minus the total ionization energy deposited in the layer, and v is the total number of displacements in the layer
- scale: what type of scale to use for first y-axis. Optional, defaults to 'linear'.
  - 'linear': use a linear scale
  - 'log': use a log scale
- layerdepth: an array of the bottom depths of each layer of the target material, must be given in angstrom. Optional, if not specified the layer bottom depths will be automatically read from the TDATA.txt file. layerdepth should always be manually input for multilayer targets to avoid calculation errors but does not need to be specified for single layer targets. Note that if specified, layerdepth must be an array, even if there is only 1 target material layer (must be a 1-element array in this case).

## Example Output

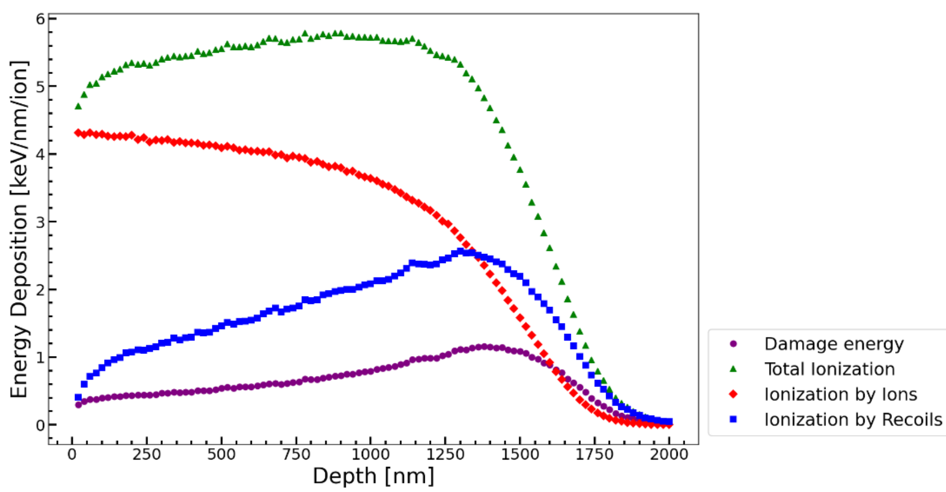



*Figure 2: Energy deposition plot for 10 MeV Au in SiC. 2000 ions calculated. Damage energy calculated using energy conservation method.*

# stopping

## Description

This function generates a plot of the stopping power profiles (nuclear and electronic). The electronic to nuclear stopping power ratio, or the reciprocal ratio, can also be plotted using a second y-axis. The first y-axis can use a linear or logarithmic scale (base 10).

## Unique Inputs

- ratio: indicator for what stopping power ratio to plot. Optional, defaults to None
  - 'e/n': plot the electronic to nuclear stopping power ratio on a second y-axis
  - 'n/e': plot the nuclear to electronic stopping power ratio on a second y-axis
  - None: do not plot either ratio
- scale: what type of scale to use for first y-axis. Optional, defaults to 'linear'.
  - 'linear': use a linear scale
  - 'log': use a log scale

## Example Output

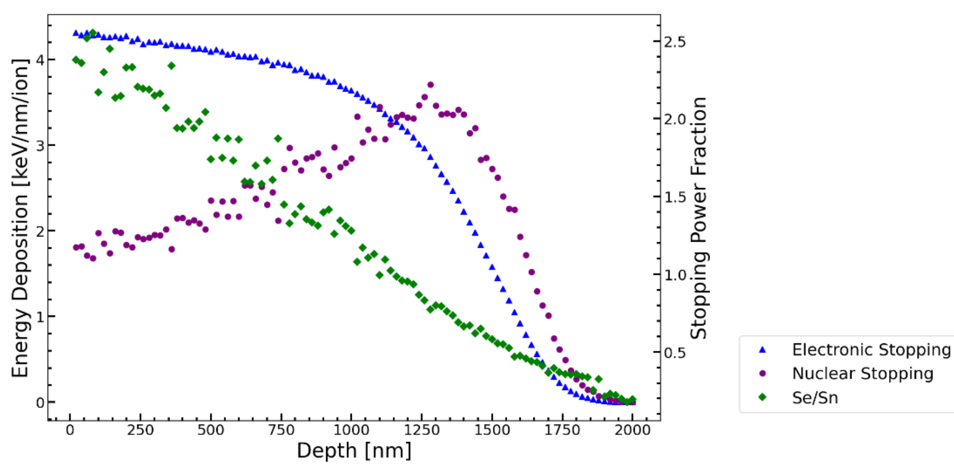



*Figure 3: Stopping power plot for 10 MeV Au in SiC. 2000 ions calculated.*

# TdamCompare

## Description

This function generates a plot of the damage energy profile calculated using each of the three methods (Phonon Method 1, Phonon Method 2, Energy Conservation Method) and prints the total damage energy calculated using each method (to console). Phonon Method 1 calculates the damage energy profile by summing the *phonons by ion* and *phonons by recoils* profiles from the PHONON.txt file. Phonon Method 2 calculates the damage energy profile only using *phonons by recoils* from the PHONON.txt file. Energy Conservation Method calculates the damage energy profile by multiplying each layer of the total displacement profile by a local Tdam/v factor specific to that layer, where Tdam is the damage energy calculated as incident ion energy deposited in the layer minus the total ionization energy deposited in the layer, and v is the total number of displacements in the layer.

### Unique Inputs

- layerdepth: an array of the bottom depths of each layer of the target material, must be given in angstrom. Optional, if not specified the layer bottom depths will be automatically read from the TDATA.txt file. layerdepth should always be manually input for multilayer targets to avoid calculation errors but does not need to be specified for single layer targets. Note that if specified, layerdepth must be an array, even if there is only 1 target material layer (must be a 1-element array in this case).

### Example Output

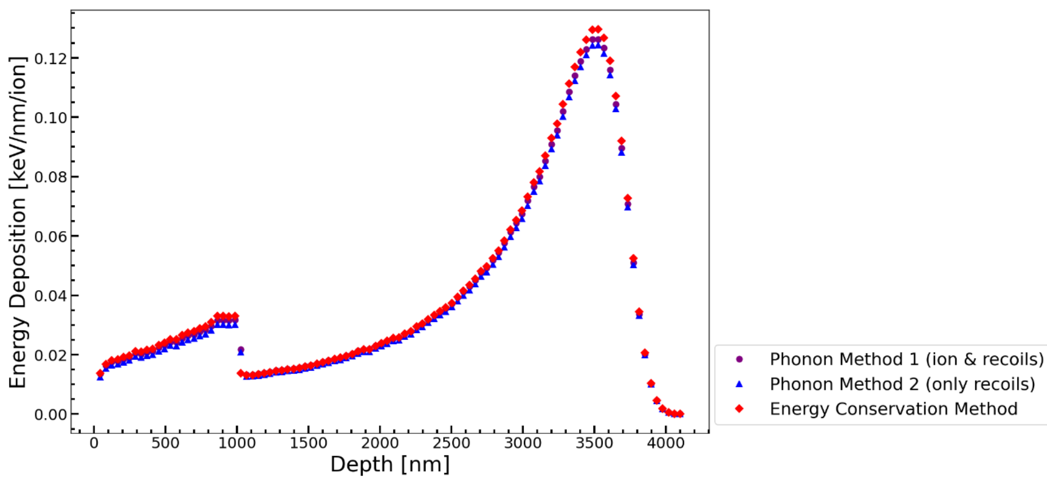



*Figure 4: Damage energy method comparison plot 9 MeV Si in ZrN (1000 nm) and Si (3100 nm). 20000 ions calculated.*

# Advanced Plotting Functions

These functions automatically extract data from the SRIM output files, then perform calculations and generate publication-quality plots that are less commonly used than the basic plots. These functions have various optional inputs, and plotting parameters (i.e. line width, colours, etc.) can be easily edited within the functions to allow for plot customization.

## collisionPlotDepth

### Description

This function generates a contour graph of the distribution of collisions within the target material, projected onto either the xy or xz plane (where x is the depth into the target). A second plot of the collision distribution projected onto the x-axis is also generated. Either the distribution of PKA collisions (ion-target collisions) or all collisions (ion-target collisions and target atom recoils cascades) can be plotted. The number of ions used for the plot can be controlled, and the colour bar can be set to either linear or logarithmic (base 10). The number of ions used is printed to console.

## Important Notes

To use this function, the “collision details” option must be manually enabled in the TRIM setup window before running the simulation, shown in the image below. When prompted after enabling this option, data must be saved for recoils as well, even if you only want to plot the PKA collisions. Then “path” must contain the COLLISON.txt file generated. Additionally, the SRIM simulation you want to use the data from has to run uninterrupted until at least the number of ions you want to use for the plot has been calculated. This means that the SRIM window cannot be closed, then resumed from disk or SRIM directory until this time. However, the SRIM simulation can still be paused.

Generating the COLLISON.txt output file requires SRIM to record details of every collision for all ions throughout the simulation. This can significantly slow down the simulation, so it is recommended to only enable the “collision details” option if the srimpro functions which use the COLLISON.txt file are to be used after the simulation.

The amount of time required to generate this plot greatly depends on the number of collisions per ion and number of ions used. For the example output below, the function took 6 minutes to generate the plot.


## Unique Inputs

- n1: number of depth bins to use (higher value will generate higher resolution plot but take more time to generate). Optional, defaults to 100.
- n2: number of bins for the second dimension (higher value will generate higher resolution plot but take more time to generate). Optional, defaults to the value of n1.
- d2: what second dimension to plot (other than x). Optional, defaults to ‘y’.
  - ‘y’: plot y as the second dimension
  - ‘z’: plot z as the second dimension
- r: the radius from the origin to plot for the y/z axis. Optional, defaults to the maximum distance of a collision between x1 and x2 from the origin (to include all collisions between x1 and x2).
- num_ions: the number of ions to use for the plot. Optional, defaults to using all ions calculated by SRIM.
- ion_start: the index of the first ion to be used for the plot; no ions before ion_start will be used. Optional, defaults to 0 (the first ion calculated).
- coll_type: what type of collision to plot. Optional, defaults to ‘ions’.
  - ‘ions’: plot only PKA (ion-target) collisions
  - ‘recoils’ plot all target atom recoil cascades, including PKA collisions

- scale: what type of scale to use for the colour bar. Optional, defaults to ‘linear’.
    - ‘linear’: use a linear scale for the colour bar
    - ‘log’: use a logarithmic scale for the colour bar

## Unique Assumptions

- n1, n2, num_ions, and ion_start are all given as integers
- r is given as a scalar

## Example Outputs

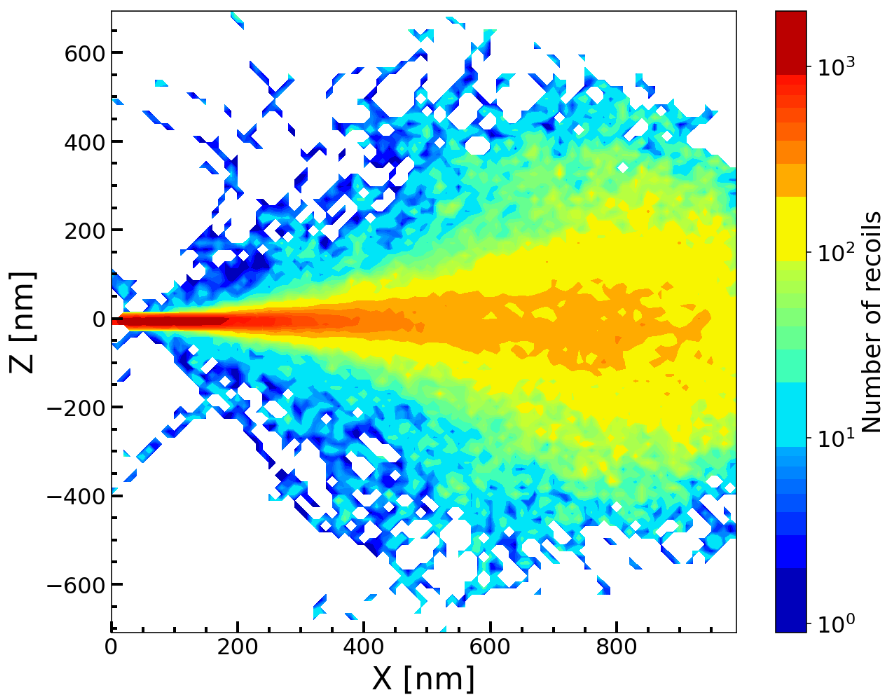



*Figure 5: Collision depth distribution plot for 1.5 MeV Si in ZrN, showing only PKA collisions. 2000 ions used, log scale used, all other options left as their defaults.*

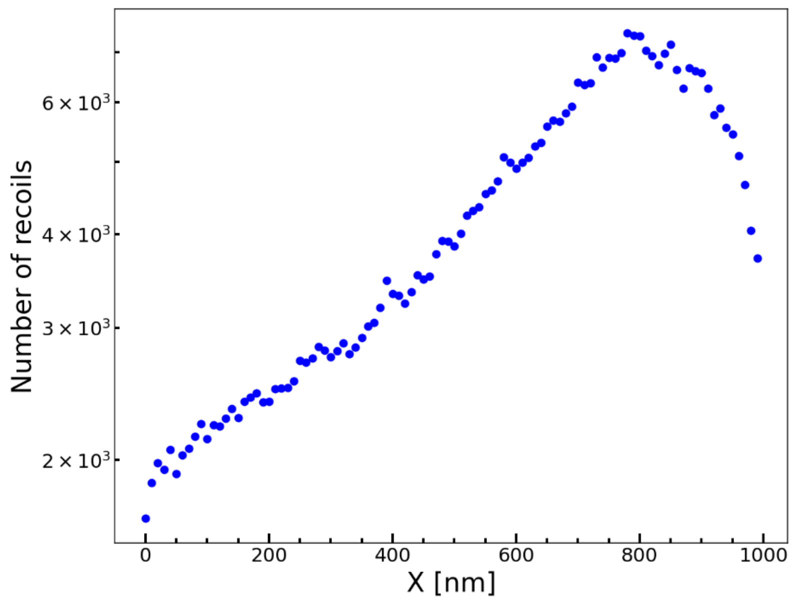



*Figure 6: Collision depth distribution (x-axis projection) for 1.5 MeV Si in ZrN, showing only PKA collisions. 2000 ions used, log scale used, all other options left as their defaults.*

## collisionPlotXsection

### Description

This function generates a contour graph of the distribution of collisions within the target material, projected onto the yz plane (where x is the depth into the target). Either the distribution of PKA collisions (ion-target collisions) or all collisions (ion-target collisions and target atom recoils cascades) can be plotted. The number of ions used for the plot can be controlled, and the colour bar can be set to either linear or logarithmic (base 10). The number of ions used is printed to console.

### Important Notes

To use this function, the “collision details” option must be manually enabled in the TRIM setup window before running the simulation, shown in the image below. When prompted after enabling this option, data must be saved for recoils as well, even if you only want to plot the PKA collisions. Then “path” must contain the COLLISON.txt file generated. Additionally, the SRIM simulation you want to use the data from must run uninterrupted until at least the number of ions you want to use for the plot has been calculated. This means that the SRIM window cannot be closed, then resumed from disk or SRIM directory until this time. However, the SRIM simulation can still be paused.

Generating the COLLISON.txt output file requires SRIM to record details of every collision for all ions throughout the simulation. This can significantly slow down the simulation, so it is recommended to only enable the “collision details” option if the srimpro functions which use the COLLISON.txt file are to be used after the simulation.

The amount of time required to generate this plot greatly depends on the number of collisions per ion and number of ions used. For the example output below, the function took 3-10 minutes to generate the plot.


## Unique Inputs

- n: number of bins to use for the y and z dimensions (higher value will generate higher resolution plot but take more time to generate). Optional, defaults to 100.
- r: the radius from the origin to plot for the y/z axes. Optional, defaults to the maximum distance of a collision between x1 and x2 from the origin (to include all collisions between x1 and x2).
- num_ions: the number of ions to use for the plot. Optional, defaults to using all ions calculated by SRIM.
- ion_start: the index of the first ion to be used for the plot; no ions before ion_start will be used. Optional, defaults to 0 (the first ion calculated).
- coll_type: what type of collision to plot. Optional, defaults to 'ions'.
    - 'ions': plot only PKA (ion-target) collisions
    - 'recoils' plot all target atom recoil cascades, including PKA collisions
- scale: what type of scale to use for the colour bar. Optional, defaults to 'linear'.
    - 'linear': use a linear scale for the colour bar
    - 'log': use a logarithmic scale for the colour bar

## Unique Assumptions

- n, num_ions, and ion_start are all given as integers
- r is given as a scalar

## Example Output

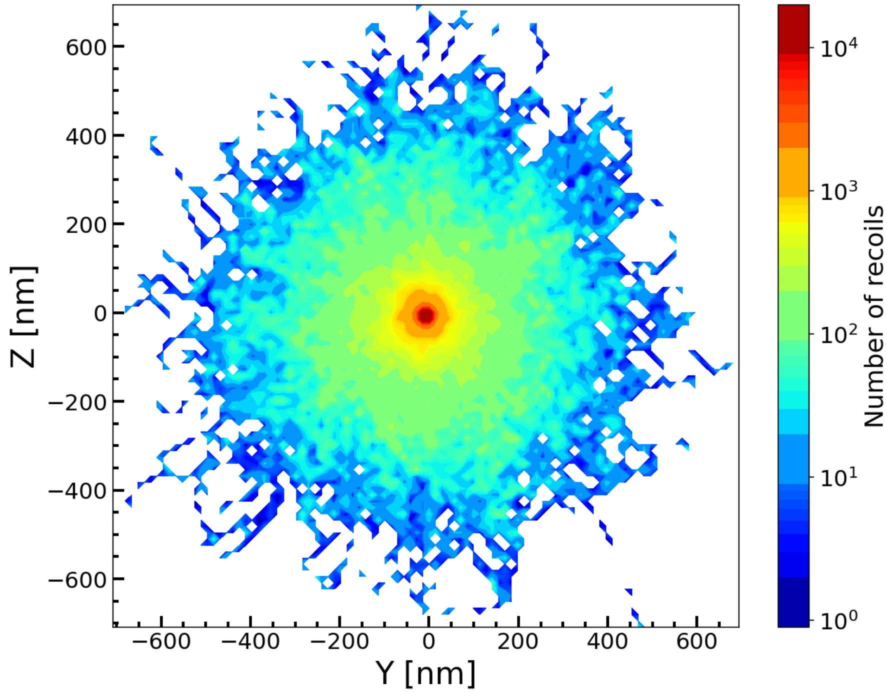



*Figure 7: Collision distribution plot for 1.5 MeV Si in ZrN, showing only PKA collisions. 2000 ions used, log scale used, only showing collisions up to 1000 nm depth, all other options left as their defaults.*

# multiRangeAndDpa

## Description

This function generates two plots, one plot of ion distributions, and one plot of damage doses, for data from multiple SRIM simulations at once (all on the same two plots). A reference layer can be specified, meaning what layer for which its start is set as the 0-depth reference point. No data will be plotted prior to the reference layer. The ion distributions and damage doses can be plotted as individual lines or combined into total ion distribution and damage dose profiles. Ion distribution can be plotted in units of [at. %] or [(Atoms/cm^3)/(Atoms/cm^2)].

## Unique Inputs

- paths: instead of passing in a single path, an array of paths must be passed in, where each element is a path to a folder containing all the SRIM output files for a simulation. Must be an array even if only 1 path is passed to the function (a 1 element array is required in this case).
- labels: an array of labels for each set of SRIM output data to be plotted, where each element is a string. The size of labels must be the exact same as that of paths, and it must be an array even if only 1 path is passed to the function.
- fluence: the fluence to use for the damage dose calculations. Must be given in $cm^{-2}$.
- units: what units to use for ion density. Optional, defaults to 'at_%'.

- 'at_%': use ion density units of [at. %]
- 'density_per_fluence': use ion density units of [(Atoms/cm^3)/(Atoms/cm^2)]

- plot_style: how to plot the data from each path. Optional, defaults to 'individual'.
  - 'individual': the ion distribution and damage dose profiles from each path are plotted as individual lines (still on the same plots)
  - 'combined': the ion distribution and damage dose profiles from each path are combined into single profiles. This option also prints the mean ion density and damage dose, and their associated errors (between x1 and x2) to console. These values are calculated both using all points in the combined profiles and only using the maximum and minimum points of the combined profiles (all values are printed to console).
- ref_layer: what layer to use as the reference layer. Optional, defaults to 1 (the first layer).
- layerdepth: an object array where each element corresponds to a path and is an array with the bottom depths of each target material layer for that path. Layer depths must be given in angstrom. Optional, if not specified the layer bottom depths will be automatically read from the TDATA.txt files. It is recommended to always manually input layerdepth if any of the SRIM simulations to be used have multilayer targets.

## Unique Assumptions

- fluence is given as a scalar
- ref_layer is given as an integer

## Example Outputs

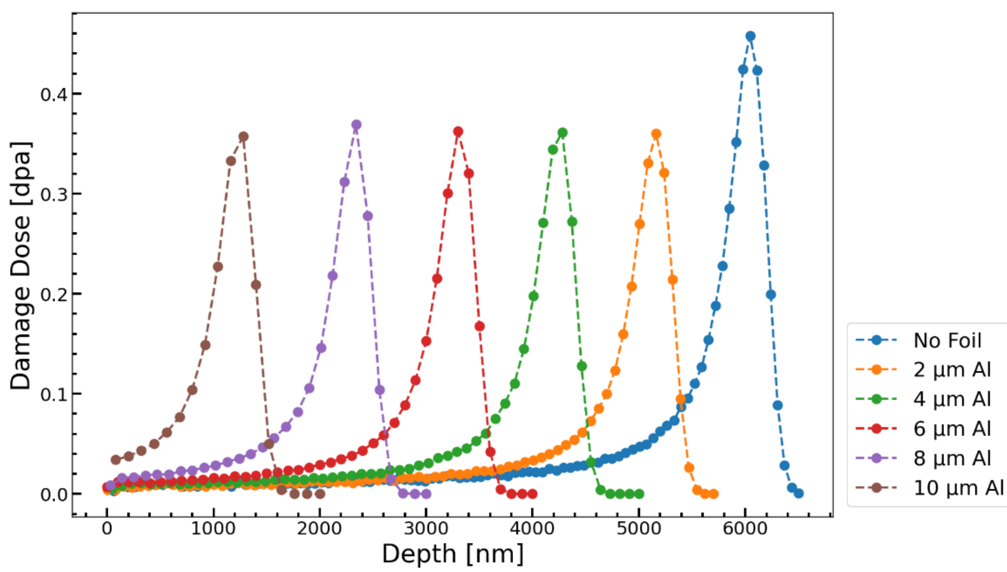



*Figure 8: Damage dose plot for different thicknesses of an Al foil on a Cu layer under 3.25 MeV He irradiation. Fluence = 6.7e15.*

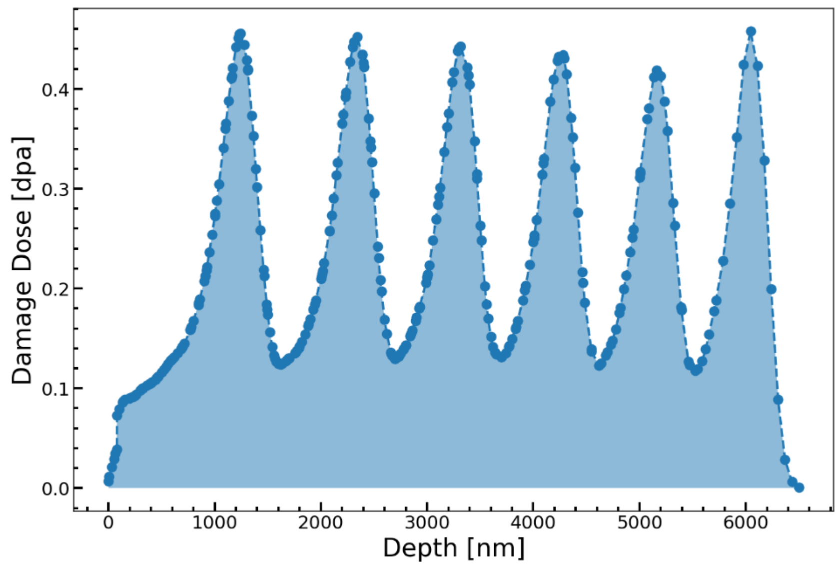



*Figure 9: Combined damage dose plot for different thicknesses of an Al foil on a Cu layer under 3.25 MeV He irradiation. Fluence = 6.7e15, reference layer = 2 (depth refers to depth into Cu layer).*

# recoilSpectra

## Description

This function generates a plot of the PKA energy spectra (fraction of total PKA recoils at or up to various energy levels/bins) for multiple SRIM simulations (all on the same plot) or exports the data to excel. Either the total or cumulative spectra can be calculated and plotted. The default options for all inputs are set to generate a plot in the most common form, producing cumulative spectra. The maximum and minimum energy to plot can be specified, and the number of ions to use can be controlled. The number of ions used for each path, and the fraction of the total number of recoils that is shown in the main plot are printed to console. Note that for both plots, the Fraction of PKA Recoils axis refers to all PKA recoils, not just those within the specified energy range.

## Important Notes

To use this function, the “collision details” option must be manually enabled in the TRIM setup window before running the simulation, shown in the image below. When prompted after enabling this option, data must be saved for recoils as well, even if you only want to plot the PKA collisions. Then “path” must contain the COLLISON.txt file generated. Additionally, the SRIM simulation you want to use the data from must run uninterrupted until at least the number of ions you want to use for the plot has been calculated. This means that the SRIM window cannot be closed, then resumed from disk or SRIM directory until this time. However, the SRIM simulation can still be paused.

Generating the COLLISON.txt output file requires SRIM to record details of every collision for all ions throughout the simulation. This can significantly slow down the simulation, so it is recommended to only enable the “collision details” option if the srimpro functions which use the COLLISON.txt file are to be used after the simulation.

The amount of time required to generate this plot greatly depends on the number of collisions per ion, the number of ions used, and the number of SRIM simulations to plotted. For the example output below, the function took 13 minutes to generate the plot.


## Unique Inputs

- paths: instead of passing in a single path, an array of paths must be passed in. Every element must be a file path to a folder containing all the SRIM output files for a simulation, or every element must be a file path to an excel file. If excel files are used, each must only contain data for a single primary recoil spectrum. Each excel file must follow specific formatting: the index 0 row must be the header row (no other header row), the index 0 column must contain the upper bounds of the energy bins (in eV) and the index 1 column must contain the non-weighted spectrum data (cumulative or total). Note that this is the same formatting produced by the 'excel' output option. paths must be an array even if only 1 path is passed to the function (a 1-element array is required in this case).
- labels: an array of labels for each set of SRIM output data to be plotted, where each element is a string. The size of labels must be the exact same as that of paths, and it must be an array even if only 1 path is passed into the function (a 1-element array is required in this case).
- n: number of energy bins to use (higher value will generate higher resolution plot but take more time to generate). Optional, defaults to 100.
- num_ions: the number of ions to use for the plot. Optional, defaults to using all ions calculated by SRIM.
- ion_start: the index of the first ion to be used for the plot; no ions before ion_start will be used. Optional, defaults to 0 (the first ion calculated).
- minE: minimum energy to plot. Optional, defaults to the lowest energy of all PKA collisions within paths.
- maxE: maximum energy to plot. Optional, defaults to the highest energy of all PKA collisions within paths.
- spectra: what type of spectra to calculate and plot. Optional, defaults to 'cumulative'.
  - 'cumulative': the most common type of recoil spectrum plot. The y-axis corresponds to the fraction of the total number of PKA recoils that occurred at or below the energy given on the x-axis
  - 'total': the y-axis corresponds to the fraction of the total number of PKA recoils that occurred at the energy given on the x-axis (within the corresponding energy bin)
- xscale: what scale to use for the plot x-axis and energy binning. Optional, defaults to 'log'.
  - 'log': use a log base 10 scale

- ‘linear’: use a linear scale
- yscale: what scale to use for the plot y-axis. Optional, defaults to ‘linear’.
  - Same options as xscale
- graph_style: how to plot the recoil spectra. Optional, defaults to ‘line’.
  - ‘bar’: recoil spectra plotted as a histogram-like bar graph
  - ‘line’: recoil spectra plotted as a normal line graph
- full_range: indicator for whether to generate a second plot of the full energy range of the PKA energy spectra. Optional, defaults to False.
  - True: enable this option
  - False: do not enable this option
- excelpath: path to an additional excel file with data to plot. Optional, defaults to not plotting additional data. Note that for this option, it is assumed that the data within the excel file is of the same form as the spectra option selected (i.e. if spectra=cumulative it is assumed that the excel file contains cumulative spectrum data). This file requires specific formatting: the index 0 row must be the header row (no other header row), the index 0 column must contain the upper bounds of the energy bins (in eV), and the index 1 column must contain the non-weighted spectrum data (cumulative or total fraction). If an additional excel file is used, a label for its data must be included as the last element of labels (labels should be 1 element longer than paths in this case).
- output: what output to get from the function. Optional, defaults to ‘plot’.
  - ‘plot’: function generates a plot
  - ‘excel’: function exports the calculated data to excel. A separate excel file for corresponding to the data from each path is created. Note that this output option will not work if the files given in paths are excel files.
- x1 and x2: this function does not have the x1 nor x2 optional inputs.

## Unique Assumptions

- n, num_ions, and ion_start are all given as integers

Example Output

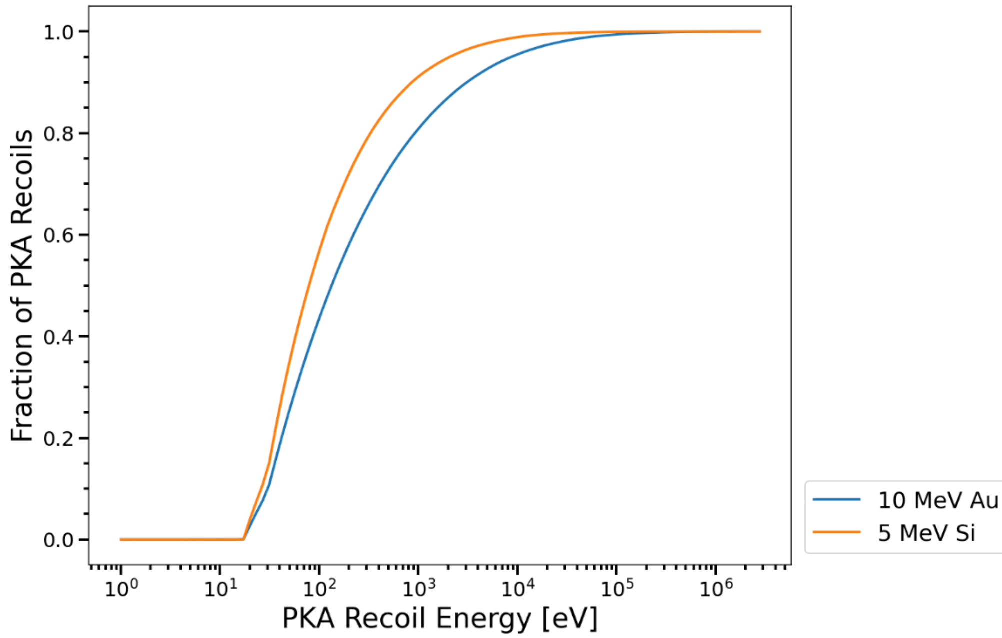



*Figure 10: Cumulative PKA energy spectra for 10 MeV Au and 5 MeV Si in SiC, 530 ions used for both. All other setting left as their defaults.*

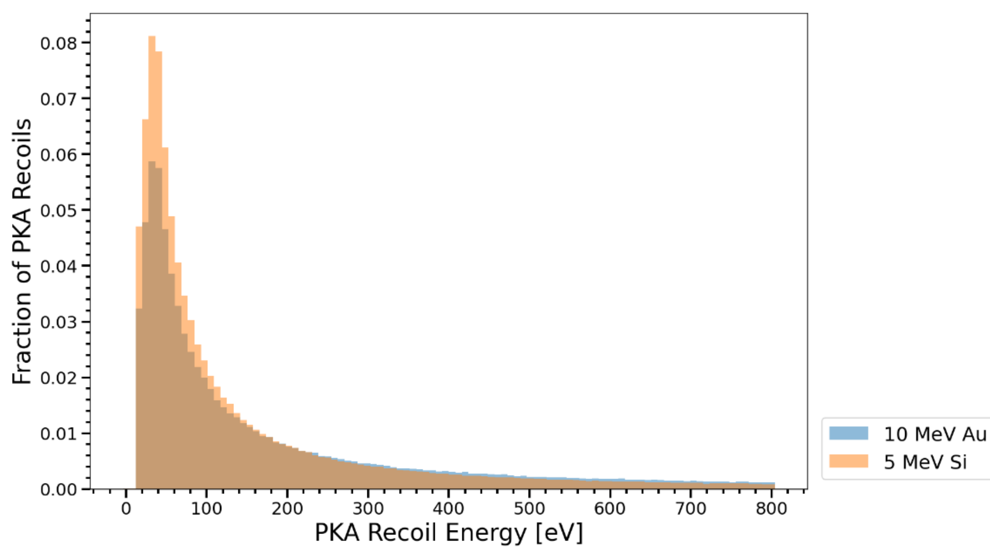



*Figure 11: Total PKA energy spectra for 10 MeV Au and 5 MeV Si in SiC. 2000 ions used for 5 MeV Si and 534 ions used for 10 MeV Au. Minimum energy = 0 eV, maximum energy = 800 eV, 'bar' graph style used, 'total' spectra option used.*

## weightedRecoilSpectra

### Description

This function generates a plot of the weighted PKA energy spectra (fraction of total PKA recoils at or up to various energy levels/bins, weighted by the number of displacements caused by PKAs at or up to that energy level/bin) for multiple SRIM simulations (all on the same plot), or exports the data to excel. Either the total or cumulative spectra can be calculated and plotted. The default options for all inputs are set to generate a plot in the most common form, producing cumulative spectra. The maximum and minimum energy to plot can be specified, and the number of ions to use can be controlled. The number of ions used for each path, and the fraction of the total number of recoils that is shown in the main plot are printed to console. Note that for both plots, the Fraction of PKA Recoils axis refers to all PKA recoils, not just those within the specified energy range.

### Important Notes

To use this function, the “collision details” option must be manually enabled in the TRIM setup window before running the simulation, shown in the image below. When prompted after enabling this option, data must be saved for recoils as well, even if you only want to plot the PKA collisions. Then “path” must contain the COLLISON.txt file generated. Additionally, the SRIM simulation you want to use the data from must run uninterrupted until at least the number of ions you want to use for the plot has been calculated. This means that the SRIM window cannot be closed, then resumed from disk or SRIM directory until this time. However, the SRIM simulation can still be paused.

Generating the COLLISON.txt output file requires SRIM to record details of every collision for all ions throughout the simulation. This can significantly slow down the simulation, so it is recommended to only enable the “collision details” option if the srimpro functions which use the COLLISON.txt file are to be used after the simulation.

The amount of time required to generate this plot greatly depends on the number of collisions per ion, the number of ions used, and the number of SRIM simulations to plotted. For the example output below, the function took 13 minutes to generate the plot.

## Unique Inputs

- paths: instead of passing in a single path, an array of paths must be passed in. Every element must be a file path to a folder containing all the SRIM output files for a simulation, or every element must be a file path to an excel file. If excel files are used, each must only contain data for a single primary recoil spectrum. Each excel file must follow specific formatting: the index 0 row must be the header row (no other header row), the index 0 column must contain the upper bounds of the energy bins (in eV) and the index 1 column must contain the weighted spectrum data (cumulative or total). Note that this is the same formatting produced by the 'excel' output option. paths must be an array even if only 1 path is passed to the function (a 1-element array is required in this case).
- labels: an array of labels for each set of SRIM output data to be plotted, where each element is a string. The size of labels must be the exact same as that of paths, and it must be an array even if only 1 path is passed into the function (a 1-element array is required in this case).
- n: number of energy bins to use (higher value will generate higher resolution plot but take more time to generate). Optional, defaults to 100.
- num_ions: the number of ions to use for the plot. Optional, defaults to using all ions calculated by SRIM.
- ion_start: the index of the first ion to be used for the plot; no ions before ion_start will be used. Optional, defaults to 0 (the first ion calculated).
- minE: minimum energy to plot. Optional, defaults to the lowest energy of all PKA collisions within paths.
- maxE: maximum energy to plot. Optional, defaults to the highest energy of all PKA collisions within paths.
- spectra: what type of spectra to calculate and plot. Optional, defaults to 'cumulative'.
    - 'cumulative': the most common type of recoil spectrum plot. The y-axis corresponds to the fraction of the total number of PKA recoils that occurred at or below the energy given on the x-axis
    - 'total': the y-axis corresponds to the fraction of the total number of PKA recoils that occurred at the energy given on the x-axis (within the corresponding energy bin)
- xscale: what scale to use for the plot x-axis and energy binning. Optional, defaults to 'log'.
    - 'log': use a log base 10 scale
    - 'linear': use a linear scale
- yscale: what scale to use for the plot y-axis. Optional, defaults to 'linear'.
    - Same options as xscale
- graph_style: how to plot the recoil spectra. Optional, defaults to 'line'.
    - 'bar': recoil spectra plotted as a histogram-like bar graph
    - 'line': recoil spectra plotted as normal line graph
- excelpath: path to an additional excel file with data to plot. Optional, defaults to not plotting additional data. Note that for this option, it is assumed that the data within the excel file is of the same form as the spectra option selected (i.e. if spectra=cumulative it is assumed that the excel file contains cumulative spectrum data). This file requires specific formatting, see the excel_format input for how to format this file. If an additional excel file is used, a label for its data must be included as the last element of labels (labels should be 1 element longer than paths in this case).

- output: what output to get from the function. Optional, defaults to ‘plot’.
    - ‘plot’: function generates a plot
    - ‘excel’: function exports the calculated data to excel. A separate excel file for corresponding to the data from each path is created. Note that this output option will not work if the files given in paths are excel files.
- excel_format: the format of the excel file given in excelpath. Optional, defaults to 1.
    - 1: the index 0 row must be the header row (no other header row), the index 0 column must contain the upper bounds of the energy bins (in eV), and the index 1 column must contain the weighted spectrum data (cumulative or total fraction). Note that this si the same format produced by the ‘excel’ output option.
    - 2: the index 0 row must be the header row (no other header row), the index 0 column must contain the upper bounds of the energy bins (in eV), the index 1 column must contain the unweighted spectrum data (cumulative or total fraction), and the index 2 column must contain either the displacement or damage energy data for each energy bin
- x1 and x2: this function does not have the x1 nor x2 optional inputs.

## Unique Assumptions

- n, num_ions, and ion_start are all given as integers

## Example Output

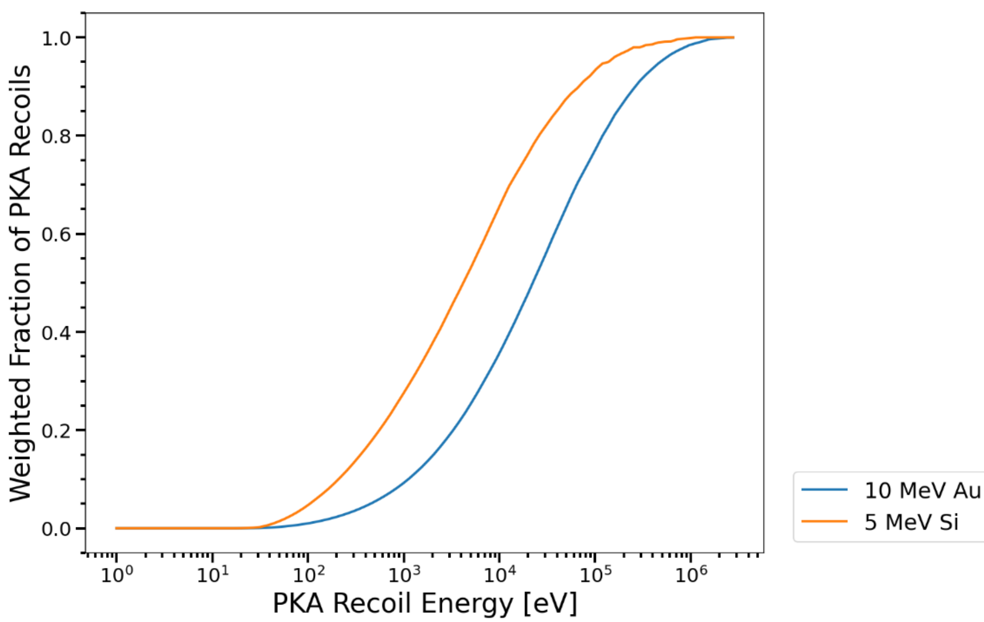



*Figure 12: Weighted cumulative PKA energy spectra for 10 MeV Au and 5 MeV Si in SiC, 530 ions used for both. All other setting left as their defaults.*

# Notes About SRIM Data Recording

Unreliable layer depths/widths recorded in TDATA.txt

- Any function in this library that uses the depth or density of the target layers for calculations gets that data from the TDATA.txt file.
- SRIM records bottom depth values for each layer in this file, but only to one significant figure (rounds to the nearest highest order of magnitude term).
- This will cause incorrect calculations for things like damage dose, ion density, and damage energy profile if the actual target layer bottom depths have more than one significant figure
- All relevant functions have a feature that overrides this error with the actual correct value for single layer targets, but no such correction is possible for multilayer targets.
- It is important to recognize this and manually pass in the bottom layer depths to such functions in these cases (when multilayer targets are present). Functions where this is significant:
    - rangeAndDpa
    - energyDep
    - TdamCompare
    - multiRangeAndDpa

Inconsistent formatting of COLLISON.txt file

- This SRIM output file is called "COLLISON.txt" not "COLLISION.txt", this is a typo from SRIM not from srimpro
- To use the collision distribution functions in this library, the collision details option must be enabled prior to running the SRIM simulation.
- If the SRIM window is closed while the simulation is running and saved, the resumed from the .sav files later, SRIM will record all subsequent collision details in a different format in the COLLISON.txt file.
- The pysrim library that srimpro uses to read the SRIM output files cannot read this new format, which is why the simulation cannot be saved then resumed from save if the collision distribution functions are to be used.
- This does not apply to any other functions (can freely save then resume from saved without corrupting any output files other than the COLLISON.txt file).